Module 1

Real Number System and **Set Notations**.

Real Number System

The real number system encompasses all numbers that can be represented on the number line. It is divided into two main groups: **Rational Numbers** and **Irrational Numbers**.

Examples of Real Numbers:

1. **Rational Numbers**:
   - $\frac{3}{4}$ (can be written as a fraction)
   - -2.5 (terminating decimal)
   - $0.6666\ldots$ (repeating decimal, equivalent to $\frac{2}{3}$)
   - 7 (whole number, also rational)
   - $-\frac{10}{5} = -2$ (simplified fraction)
2. **Irrational Numbers**:
   - π (non-repeating, non-terminating decimal)
   - $\sqrt{2}$ (cannot be expressed as a fraction)
   - e (Euler's number, approximately 2.71828...)
   - $\sqrt{3}$ (irrational square root)
   - $0.1010010001\ldots$ (non-repeating, non-terminating pattern)

Natural Numbers (Counting Numbers)

Natural numbers are the positive integers starting from 1 and extending infinitely. They are used for counting objects.

Examples of Natural Numbers:

1. 1 (smallest natural number)
2. 5 (used for counting five objects)
3. 100 (a larger natural number)
4. 2023 (current year, a natural number)
5. 999 (natural number close to 1000)

Whole Numbers

Whole numbers include all natural numbers along with zero. They represent quantities without fractions or decimals.

Examples of Whole Numbers:

1. 0 (represents nothing or null value)
2. 1 (smallest positive whole number)
3. 10 (a small whole number)
4. 50 (used in counting or measurement)
5. 1000 (a large whole number)

Integers

Integers include all whole numbers and their additive inverses (negative counterparts). This set contains positive numbers, negative numbers, and zero.

Examples of Integers:

1. -5 (negative integer)
2. 0 (neutral integer)
3. 3 (positive integer)
4. -100 (large negative integer)
5. 2023 (large positive integer)

Rational Numbers

Rational numbers are those that can be expressed as a ratio of two integers ($\frac{a}{b}$), where $b \neq 0$. These include terminating and repeating decimals.

Examples of Rational Numbers:

1. $\frac{1}{2} = 0.5$ (terminating decimal)
2. $-\frac{3}{4} = -0.75$ (negative rational number)
3. $0.3333... = \frac{1}{3}$ (repeating decimal)
4. $7 = \frac{7}{1}$ (integer, also rational)
5. $-2.25 = -\frac{9}{4}$ (negative rational number)

Irrational Numbers

Irrational numbers cannot be expressed as a ratio of two integers. Their decimal expansions are non-repeating and non-terminating.

Examples of Irrational Numbers:

1. $\pi \approx 3.14159...$ (famous irrational constant)
2. $\sqrt{2} \approx 1.41421...$ (square root of 2)
3. $e \approx 2.71828...$ (base of natural logarithms)
4. $\sqrt{5} \approx 2.23606...$ (another irrational square root)
5. $0.1234567891011...$ (non-repeating, non-terminating decimal)

Even and Odd Numbers

Even numbers are divisible by 2, while odd numbers are not. Both are subsets of integers.

Examples of Even Numbers:

1. 2 (smallest positive even number)
2. -4 (negative even number)
3. 10 (even number greater than 5)
4. 100 (large even number)
5. 0 (neutral even number)

Examples of Odd Numbers:

1. 1 (smallest positive odd number)
2. -3 (negative odd number)
3. 7 (odd number greater than 5)
4. 99 (large odd number)
5. -11 (negative odd number)

Prime and Composite Numbers

Prime numbers are integers greater than 1 that have no divisors other than 1 and themselves. Composite numbers have more than two divisors.

Examples of Prime Numbers:

1. 2 (only even prime number)
2. 3 (smallest odd prime number)
3. 5 (prime number greater than 3)
4. 13 (larger prime number)
5. 29 (prime number less than 30)

Examples of Composite Numbers:

1. 4 (divisible by 1, 2, and 4)
2. 6 (divisible by 1, 2, 3, and 6)
3. 8 (divisible by 1, 2, 4, and 8)
4. 15 (divisible by 1, 3, 5, and 15)
5. 20 (divisible by 1, 2, 4, 5, 10, and 20)

Summation Notation

Summation notation (or sigma notation) provides a compact way to represent the sum of a sequence of terms.

The general form is:

$$\sum_{i=a}^{b} f(i)$$

Where:

- i: Index variable.
- a: Starting value of the index.
- b: Ending value of the index.
- $f(i)$: Expression defining the terms to be summed.

For example:

$$\sum_{n=1}^{3}(2n-1) = [2(1)-1] + [2(2)-1] + [2(3)-1] = 1 + 3 + 5 = 9$$

Product Notation

Product notation (or pi notation) is used to represent the product of a sequence of terms. The general form is:

$$\prod_{i=a}^{b} f(i)$$

Where:

- i: Index variable.
- a: Starting value of the index.
- b: Ending value of the index.
- $f(i)$: Expression defining the terms to be multiplied.

For example:

$$\prod_{k=1}^{4} k = 1 \cdot 2 \cdot 3 \cdot 4 = 24$$

Factorial Notation

Factorial notation ($n!$) represents the product of all positive integers from 1 to n. For example:

$$n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 1$$

Special cases:

- $0! = 1$ (by definition).
- Recursive property: $n! = n \cdot (n-1)!$.

The Successor Operation

The idea is:

- **Successor** means the next number
- $S(n) = n + 1$
- Addition is built using these two rules:
  - $m + 0 = m$
  - $m + S(n) = S(m) + n$

So instead of adding directly, the program keeps:

- increasing the first number by 1
- decreasing the second number by 1
- until the second number becomes 0

Simple algorithm in numbered form

Input:

Two natural numbers `m` and `n`

Process:

1. Read `m` and `n`
2. Define successor as `S(n) = n + 1`
3. Call recursive function `add_successor(m, n)`
4. If `n == 0`, return `m`
5. Otherwise:
   - replace `m` with `S(m)`
   - replace `n` with `n - 1`
   - call the function again
6. Repeat until `n = 0`
7. Display the final result